Human Compression: the audit of a fixed corpus

Human Compression project

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1 The cutoff and the objects being compared

New manuscript intake is closed. The baseline is the Git commit

`3b067cc2eba4ac2f2f91248448727639d8ac7c5a`,

also named `intake-2026-10-11`. Its catalogue contains 161 works, of which 102 have retrieved full texts, 61 have specified encodings, 48 have a rederived result of some stated scope, and 27 have previously admitted recoveries of specified principal results. These overlapping counts do not describe 161 completed readings, 27 replacements of every assertion, or 27 fully formalized papers.

The outstanding F015 review and L028 manuscript may finish because their proof work was already underway. The bounded investigation of finite minimal covers may record proved conditional criteria and unresolved obstructions. N007 is not a further application in this intake. None of these pending items is included in the 27 admitted principal contracts. The exact cutoff records are in `audit/intake.json` and `audit/source_registry.json`.

For each source, the audit first fixes an edition and a list of results to be recovered, with all parameter ranges and exceptional cases. A correction to a false source statement is recorded as a correction, not as a proof of the original statement. Recovery of the main results does not by itself recover every auxiliary assertion, table or illustration. The source and replacement must have the same declared mathematical outputs before their lengths are compared.

2 Theory, applications and proof closure

A topic A consists of a complete shared theory U_A and complete applications $A_1, \dots, A_k$, corresponding to declared source contracts $LA_1, \dots, LA_k$. Each application must check the actual hypotheses of the shared theorems and supply its local constructions, converse arguments, exceptions, geometric identifications and any necessary exhaustive computations. Similar wording is not a proof of common structure. A paragraph saying that a theorem applies is sufficient only when all its hypotheses and the claimed output have actually been identified.

Existing proofs supply material for new common arguments. The objective is to reprove the source results as consequences of a small theory, not to shorten their wording. Moving a proof to another file earns no mathematical credit. The earlier combined manuscripts are a proof bank, not the intended replacement. The current position is as follows.

Topic	Existing material and next obligation
A	Complete shared theory and six applications, with independent mathematical and source review and a fresh full program replay. Exact source corrections and retained inputs are explicit. The complete length comparison and paired dependency audit are independently accepted.

B	Complete common theory and four applications on forests and primitive coset geometries, independently reviewed for thirty declared clauses. The length and paired dependency audits are accepted; N001's PDF cost and the numerical whole-topic factor remain unknown.
C	Complete common theory and four applications, with an explicit binary translation inverse, full geometric identifications and the fresh Janko census. Complete lengths are accepted, with L108's PDF cost unknown. The paired dependency audit is accepted: exclusive passage costs conserve all complete documents and count shared prerequisites once.
D	Complete common theory and three applications, independently reviewed with all programs and ten scoped Lean declarations. Field and group classifications remain retained inputs; L104's principal contract excludes its printed diagrams, Aut/Cor groups and doubling relation. Complete lengths and all combined/conditional fractions are accepted. The paired dependency audit conserves every document total and charges each shared passage once.
E	A new common theorem for string markings with proper exceptional ends derives the intersection property instead of assuming it. Three applications recover the two polytope classifications and the corrected PSL maniplex results. Prior editions remain separate baselines.
F	Complete common orbit, character and cyclotomic theory and the L014 application, independently reviewed. All three programs are validation; five scoped Lean declarations verify subgroup interfaces and identities. The tightness converse retains its external classification theorem. Lengths and paired dependencies remain pending.
G	Complete flag, Steiner residue and Hermitian theory, with corrected L009 and L092 applications. All 79 source records, thirteen programs and forty scoped Lean declarations reviewed. External classifications and explicit exclusions remain; lengths and paired dependencies are pending.
Affine	Complete common affine theory and corrected L012 application, including all kernel, square and cubic arguments they use. All six full programs and nine scoped Lean declarations passed independent review. Complete lengths and combined/conditional fractions are accepted; paired dependency costs remain pending.

The first publication migration supplies ten editable TeX proof texts: two in A, three in E, two in G, and three separate cover results. The newly compiled PDFs have the same extracted text and page counts as the reviewed PDFs. This checks textual preservation during conversion; it does not establish new mathematics or complete the corpus audit. The remaining proof texts and all required dependencies must undergo the same treatment before a topic is declared ready for measurement.

3 Attribution and formal verification

Each application will have a machine-readable ledger connecting a source result and its exact locator to the replacement theorem, its proof, retained inputs, required programs, and any Lean declarations. The TeX edition will distinguish borrowed results, contributions of this project, recovery of several previous results, source corrections, computational checks, formalized lemmas, and answers to previously posed questions. An assertion of historical novelty requires a separate literature audit.

For every formal claim, the audit inspects the actual Lean statement, including its hypotheses. A general transport lemma with an unproved geometric hypothesis does not formalize the geometric theorem. The ledger must name the written obligations left outside Lean. Finite computations count as formal evidence only to the extent that their certificates and the implication to the claimed theorem are checked.

The canonical library now contains 100 modules. Its build evidence combines one fresh run of 96 unchanged modules with four separately checked additions: the marked-extension module, two normal-kernel modules and the commuting retraction module. The latter has four checked definitions and seven checked theorems. No single fresh run of all 100 modules is asserted. All belong in the default build. Source hashes, toolchain, dependency revisions and axiom reports are recorded. One unchanged legacy module has four native-evaluation axioms. The generic compatible-pair inverse is formalized; the application coordinate maps and their joint faithfulness remain written obligations. Existing limited formalizations will be extended where they verify substantive steps in our own proofs. Neither successful compilation nor the number of Lean declarations measures the fraction of a paper that has been formalized.

Topic B supplies a separate freshly checked Lean module, outside that 100-module default library. Its six declarations prove the actual residual-generation induction and its subgroup consequences. The geometric adapter, weighted-forest theorem and four complete applications remain written mathematics. The public packet includes the precise statement map, standard-axiom reports and finite witnesses to the necessity of its hypotheses. No source contract has acquired an end-to-end formalization.

Topic C has a separate three-module check: nine new declarations for the explicit semidirect inverse, plus seven unchanged displacement and intersection declarations. All sixteen were freshly compiled by the independent reviewer and by root. The augmentation basis, source-specific maps and geometry, and the Janko census remain outside Lean. Its full finite census was rerun independently, using both complete required programs and the retained subgroup catalogue. The legacy-plus-B runner does not yet dispatch Topic C; its separate command is published with the edition.

The affine edition has nine separately replayed declarations: four reused quotient-intersection criteria, three reused sign-obstruction lemmas and two statements preserving intersections under smaller normal periods. Actual compiled types and hypotheses were inspected. Affine coordinates, presentation rewriting, catalogue completeness and the full geometric classification remain outside these Lean proofs. All six complete mathematical programs were independently replayed, including the full order-384 nonexistence search. This does not add an end-to-end source formalization or a new solved-question credit.

4 Length and reuse

Fix the source contracts, representations, document boundaries and length convention c before reporting a factor. The intended principal comparison uses the full source manuscripts once their declared main-result contracts have been recovered. Ancillary material need not all be reproduced, but the omitted results and any corrections must remain explicit. A comparison restricted to matched proof passages is a separate sensitivity measure; it must not silently replace the full-manuscript numerator. The two headline measures are a complete topic and each conditional result. Write

$$R_A^{\mathrm{written}} = \frac{\sum_{i=1}^{k} c(LA_i)}{c(U_A) + \sum_{i=1}^{k} c(A_i)}, \qquad r_{A,i} \mid (U_A, P_A) = \frac{c(LA_i)}{c(A_i)}.$$

The second measure assumes that the *whole* umbrella and all required programs P_A have already been acquired, including any application-specific parts of those programs. Their acquisition

costs must be displayed. Beside the written-topic factor, report the complete required-program coordinate

$$R_A^{\text{required}} = \frac{\sum_{i=1}^{k} c(LA_i)}{c(U_A) + \sum_{i=1}^{k} c(A_i) + c(P_A)}.$$

Here P_A contains required proof artifacts not already counted in the written papers. It is charged once within the topic, without hypothetical pruning. A printed program appendix and its raw source are alternative representations, not two charges for the same program. Written and program costs use compatible units: pages are never added to code bytes. Optional diagnostic programs have a separate coordinate. If any required artifact is excluded from the prepaid library in a different comparison, its actual cost must be added to the relevant application's denominator. Classical inputs and unavailable historical artifact costs remain explicit.

The complete cost of U_B is charged in topic B, even if it overlaps U_A. Conversely, U_A is not charged again to every application in topic A. Conditional application factors must display the cost of the assumed umbrella. Missing costs are unknown, not zero. The current three-application Topic E edition costs 39,303 non-whitespace characters against 71,375 in its three sources. Its complete written factor is 1.816×, or 1.523× including the complete required program. Given the whole common theory and program, the application factors are 6.957×, 11.271× and 2.018×. The earlier two-source and three-source editions remain separately bound baselines; their costs are not substituted for those of the current proofs.

Topic A now has six applications under the same rule. Its six sources cost 195,379 characters, against 49,059 for the full common theory and 78,251 for the applications. The written factor is 1.535×; the complete program adds 53,035, giving 1.083×. Given the whole common theory and program, the six factors, in the declared order, are 2.016×, 3.706×, 2.853×, 4.107×, 3.519× and 1.310×. The source-image and retained-input limitations below apply to these figures.

Topic B charges 14,119 characters for its common theory and 44,942 for all four applications, giving a complete written denominator of 59,061. The three available source PDFs cost 57,539. If the unavailable N001 PDF has cost x, the complete comparison is $(x + 57{,}539)/59{,}061$; there is no numerical overall factor. B1 remains charged. After the whole common theory has been acquired, the factors for N001, N003, N008 and L030 are unknown, 1.254×, 1.576× and 1.811×, respectively. No mathematical enumeration program is required by this proof edition.

Topic D's three sources cost 47,894 characters. Its whole common paper costs 15,632 and all three applications cost 29,855. The complete written factor is therefore $47{,}894/45{,}487 = 1.053\times$. The whole printed program appendix adds 27,070, giving $47{,}894/72{,}557 = 0.660\times$; the program cost outweighs the written saving. Using all eight complete raw programs instead gives 0.668×, an alternative representation, not an extra charge. After acquiring the whole theory and all eight programs, including the three full diagnostics, the conditional factors are 1.903×, 1.546× and 1.149×. The two complete execution drivers are separately inventoried at 10,038 characters. Historical software, external proofs, catalogue construction and graphics have unknown costs. These are comparisons of the declared available texts, not estimates of total historical effort or minimum proof complexity.

The affine topic compares one complete source, costing 32,767 characters, with a whole common paper costing 19,070 and an application costing 22,638. The complete written replacement is longer, with factor $32{,}767/41{,}708 = 0.786\times$. After acquiring the whole common paper and all six required programs, the local factor is $32{,}767/22{,}638 = 1.447\times$. The full printed program appendix adds 17,250, giving 0.556×; all six raw files give 0.561× as an alternative representation. Three complete auxiliary scripts cost 16,581 and three formal files cost 7,958 separately, outside those denominators. The principal contract, source corrections and exclusions are unchanged. Its paired dependency costs remain pending.

The primary text coordinate is extracted PDF text, normalized under a fixed protocol. Exact running page furniture is excluded, while titles, abstracts, references and other document text remain charged. Both NFC and NFKC normalizations are retained, with five length conven-

tions each: whitespace words, alphanumeric units, non-whitespace characters, characters with collapsed whitespace, and UTF-8 bytes with collapsed whitespace. Necessary font-to-Unicode calibrations are bound to actual glyph occurrences and reviewed independently. Raw extraction is retained as a sensitivity.

Some scientific diagrams and labels are images. Their absence from extraction does not make their cost zero. In particular, the L035 source in Topic A has scientific image content, including a nonblank page whose extracted text is empty. Its text factor therefore measures complete-document extractable text, not all visible content. Page counts and known image limitations are reported separately. Missing graphic, historical-program and external-proof costs remain unknown; program-inclusive factors are comparisons of known costs.

A separate sequential account follows the reuse discipline employed in the GKM project. Let K_{i-1} be the already paid-for library and D_i the full dependency closure of the next application, in an additive representation of identified proof units. Then

$$\Delta_i = c(D_i \setminus K_{i-1}), \qquad K_i = K_{i-1} \cup D_i.$$

Hypothesis checks, new interfaces and required programs belong to D_i. The fixed graph and an explicit admissible order determine this account. Without unrecorded revisions, the increments telescope to the cost of the acquired union; no earlier theory disappears from the total. This sequential account does not alter the independent topic costs.

Both the source and replacement require theorem-dependency graphs with the same output contracts. Complete proof units, including exposition and interfaces, receive costs. A citation alone does not establish that a result is used; a cited theorem that is actually used remains an explicit retained-input edge. Publish per-paper costs, cumulative costs and sequential acquisition costs. A sawtooth pattern must emerge from those measurements. It is neither an assumption of the audit nor an established conclusion of the current work.

5 Dependency nodes and their costs

For each matched result, construct a directed acyclic graph whose arrows run from a prerequisite to the argument using it. Numbered theorems, lemmas, propositions and corollaries give a first level of proof units. Substantive unnumbered arguments receive explicit units too; definitions, context, references and programs are distinguished from mathematical proof nodes. A proof node contains its full statement, proof and local hypothesis checks. Repeating a statement does not create a second result, but the printed repetition still has a cost.

For an output t, let $D(t)$ contain t and all its prerequisites. We report the number $N(t)$ of mathematical proof nodes in $D(t)$, the list of external inputs, and the known weighted cost

$$W_{\text{known}}(t) = \sum_{v \in D(t),\, w(v) \text{ known}} w(v).$$

The omitted unknown weights are listed explicitly; this sum is never presented as an unconditional total. Statements retained from a classical library have their printed statement cost and a separate unknown proof cost. Required programs are charged with the proof they support.

Each measured document is partitioned into disjoint text spans. A span has exactly one cost owner. Nodes can own several spans; node weights are their combined costs. Context outside the logical closure remains charged in the full-manuscript comparison. This makes the weighted graph additive without replacing the primary full-document numerator by a selected-proof numerator. Edges record actual use with a locator, not merely the appearance of a citation.

Node counts depend on granularity. A theorem containing six cases must not earn a sixfold saving simply because another author numbered the cases as lemmas. We therefore retain

the named-result count and expose substantive branches where bundling matters. Parent and child descriptions are two views of the same proof: costs belong to the leaves and are not charged again at their parents. The weighted total must be invariant under such regrouping. Counts, weighted costs and their granularity are reported together; no raw node ratio is called a complexity ratio.

For several outputs the acquired closure is their union, so a common proof is charged once within the topic. After acquiring U_A, the next application pays its new closure and its own hypothesis checks. Source papers receive the same reuse treatment for earlier results they actually invoke. Separate topics still pay their whole umbrellas independently, as in the preceding section. The paired graphs should identify where a new argument eliminates a chain of old arguments, where it merely moves one, and where an unchanged external theorem is still required.

6 Questions and publication

Answers to open questions are assessed separately from recovery of known results. Each entry states the exact question and edition, the answer's scope, proof dependencies, computational and formal evidence, and limits of the priority search. They are not inserted into a historical-length numerator. In particular an absent historical proof has no measured length, so an infinite compression factor is not an empirical length ratio. The project may identify a qualitative gain beyond recovery while keeping this numerical distinction explicit.

TeX and Lean are the editable mathematical sources; PDFs are readers and JSON supplies ledgers. Markdown stays in local working directories. The 992 historical Markdown files remain bound to their exact Git objects, with their paths and hashes in `audit/archive_markdown.json`. The current verifier ignores local notes and checks the archived bytes separately from the current tracked publication. The old replay programs run from a complete checkout of the pinned intake commit. They are not represented as fresh replays of the reorganized publication.

The matched dependency audits for Topics E, A, B, C and D are complete within their stated scope. Topic A assigns all 341 counted text spans to 337 exclusive owners; its 236 argument and reference nodes and 392 edges do not supply a ratio of node counts. All 140 document-cost coordinates agree with the complete-manuscript measurement. Topic B assigns 217 text spans to 214 exclusive owners, with 184 argument and reference nodes and 348 edges. Its eighty document-cost coordinates also agree with the complete-manuscript measurement. Topic C has 156 nodes, 249 typed arrows and 199 passage owners; all 270 document metric cells are conserved, while L108's source cost remains unknown. Topic D has 137 nodes, 284 typed arrows and 220 passage owners. Its 240 document metric cells and 116 prerequisite unions preserve the complete written and program comparisons. The current D graph records one corrected premise and two restored shared setup passages. None of these graphs supplies a compression factor from its number of nodes. The current work prepares the remaining topics from existing proofs. A topic closes only after independent mathematical review, complete source attribution, the declared Lean and computation checks, and a measured denominator containing every required component. This cutoff changes the work from acquisition to proof consolidation and assessment; it makes no new claim of recovered results, solved questions or compression.